

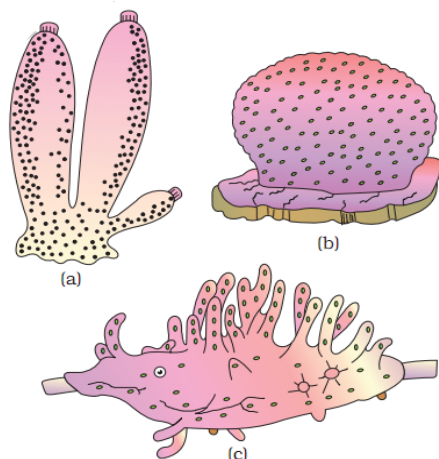


Fig 3.1 Examples for Porifera :
(a) *Sycon* (b) *Euspongia* (c) *Spongilla*

Fertilisation is internal. Cleavage is holoblastic and development is indirect involving different types of larval stages such as **parenchymula**, **trichimella** and **amphiblastula** which are morphologically different from the adult. Power of regeneration is well developed. They are considered an evolutionary blind offshoot.

Examples: *Sycon*, *Euplectella*, *Spongilla* and *Euspongia*.

3.1.2 Classification of the phylum Porifera

Phylum Porifera is classified into three classes,
1. Calcarea 2. Hexactinellida and **3. Demospongiae**

Class-1: Calcarea

They live in shallow marine waters. They are solitary or colonial. Body structure is simple. Spicules are calcareous

(made up of Ca CO_3).

Examples: *Sycon* (*Scypha*), *Leucosolenia*, *Grantia*.

Class-2: Hexactinellida

They live in deeper parts of the sea. They are solitary. Spicules are six rayed (hexactinal) and siliceous (made of silicon dioxide – *glass*).

Examples: *Euplectella* (Venus' flower basket), *Hyalonema* (Glass - rope sponge).

Class-3: Demospongiae

There are both marine and freshwater sponges in this class. They are colonial. Skeleton consists of siliceous spicules (other than six rayed) or spongin fibres or both or skeletal structures may be absent.

Examples: *Spongilla* (freshwater sponge), *Euspongia* (bath sponge), *Chalina* (dead man's fingers).

3.2 PHYLUM-CNIDARIA (COELENTERATA)

- *Tissues and the radial body plan*

3.2.1 General Characters



Cnidarians (previously called coelenterates) are aquatic, mostly marine, solitary or colonial, sessile or free-swimming, **radially symmetrical** animals (sea anemones are biradially symmetrical) (Figure 3.2). The recent name Cnidaria is derived from the '**stinging cells**' called **cnidoblasts** or **cnidocytes** present mostly on the tentacles and the body. Cnidoblasts are used for anchorage, defence and capture of prey (Figure 3.3). Cnidarians are the first metazoans to exhibit '**tissue level of organisation**' and are '**diploblastic**'. They have a central

gastro-vascular cavity or **coelenteron** (serves both digestive and circulatory functions), hence the earlier name coelenterata, with a single opening, the mouth serving the purpose of ingestion and egestion, there being no anus. Digestion is both **extracellular** (in the coelenteron) and **intracellular** (in the nutritive muscular cells of the endoderm). Nervous system is of primitive type formed of diffuse '*nerve net*' (formed by non-polarised nerve cells). Sensory structures such as **statocysts** occur in the medusoid forms. Some of the cnidarians, e.g., coral forming cnidarians have an exoskeleton composed of calcium carbonate. Cnidarians show two basic body forms called **polyp** and **medusa** (Figure 3.2). The former is a sessile and cylindrical form (e.g. *Hydra*, *Adamsia* etc.), whereas, the latter is '*umbrella-shaped*' and free-swimming form (e.g. *Aurelia* – commonly called jelly fish). Those cnidarians which exist in both forms exhibit alternation of generations (**metagenesis**), i.e., polypoid forms produce medusae asexually and medusae form the polypoid forms sexually (e.g. *Obelia*). Asexual reproduction is by budding and sexual reproduction is by syngamy. Development is indirect and includes a free swimming ciliated larva, the **planula**.

Examples: *Physalia*, *Adamsia*, *Pennatula*, *Rhizostoma* and *Meandrina*.

3.2.2 Classification of the phylum Cnidaria (Coelenterata)

Phylum Cnidaria is classified into three classes, namely **1. Hydrozoa**, **2. Scyphozoa** and **3. Anthozoa**.

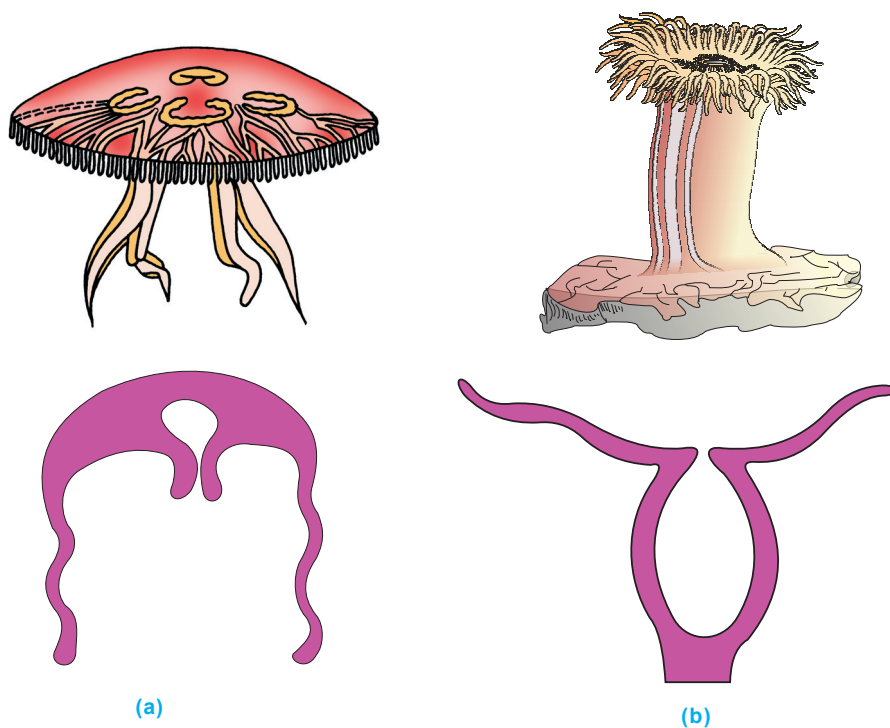


Fig. 3.2 Examples of coelenterata indicating outline of their body form:
(a) *Aurelia* (medusa) (b) *Adamsia* (polyp)